Microsoft - 100 Technical MCQ Questions

Microsoft (Software Engineer, New Grad)

For BTech Computer Science / IT Students

Object-Oriented Programming

1. Method overloading is an example of which type of polymorphism?

A) Runtime polymorphism
B) Compile-time polymorphism
C) Dynamic binding
D) Late binding

2. Which of the following is NOT a pillar of Object-Oriented Programming?

A) Encapsulation
B) Polymorphism
C) Compilation
D) Inheritance

3. What is the output-behavior difference between shallow copy and deep copy of an object?

A) Shallow copy duplicates only references to nested objects; deep copy duplicates nested objects too
B) There is no difference
C) Deep copy is always faster
D) Shallow copy always throws an exception

4. What is operator overloading?

A) Giving additional meaning to an operator for user-defined types
B) Removing operators from a language
C) A compilation error
D) A runtime exception

5. Which keyword is used in Java to prevent a class from being inherited?

A) static
B) final
C) private
D) abstract

6. In Python, which method is automatically called when an object is created?

A) __new__ then __init__
B) __del__
C) __call__
D) __str__

7. An abstract class in Java can have:

A) Only abstract methods
B) Only concrete methods
C) Both abstract and concrete methods
D) No methods at all

8. Which OOP feature allows a subclass to provide a specific implementation of a method already defined in its superclass?

A) Overloading
B) Overriding
C) Encapsulation
D) Abstraction

9. Which of these is a valid example of encapsulation in practice?

A) Making instance variables private and exposing getters/setters
B) Making all variables public
C) Using global variables everywhere
D) Avoiding classes altogether

10. What is encapsulation?

A) Hiding implementation details and exposing only functionality
B) Creating multiple classes
C) Compiling code faster
D) Using loops effectively

DBMS & SQL

11. Which isolation level allows 'dirty reads'?

A) Serializable
B) Repeatable Read
C) Read Committed
D) Read Uncommitted

12. What is the main advantage of using an ORM (Object-Relational Mapping) tool?

A) It eliminates the need for a database entirely
B) It lets developers interact with the database using objects instead of raw SQL
C) It automatically normalizes data
D) It replaces indexes

13. What is a schema in a database context?

A) The logical structure that defines tables, columns, and relationships
B) A backup file
C) A stored procedure
D) A query optimizer

14. What is the purpose of the GROUP BY clause?

A) To sort data in ascending order
B) To arrange identical data into groups for aggregate calculations
C) To filter rows before aggregation
D) To join multiple tables

15. Which of the following is an example of a NoSQL database?

A) MySQL
B) PostgreSQL
C) MongoDB
D) Oracle

16. Which SQL function is used to combine strings from multiple rows into one string (varies by RDBMS)?

A) CONCAT() only
B) GROUP_CONCAT()/STRING_AGG() depending on RDBMS
C) MERGE()
D) JOIN()

17. Which of these correctly describes a many-to-many relationship implementation?

A) Implemented directly with a foreign key in one table
B) Implemented using a junction/associative table with foreign keys to both tables
C) Not possible in relational databases
D) Requires a trigger

18. What does 'sharding' mean in database scaling?

A) Splitting a large database horizontally across multiple servers
B) Encrypting data
C) Creating a backup
D) Normalizing tables

Operating Systems

19. Which of the following is NOT one of the four necessary conditions for deadlock?

A) Mutual Exclusion
B) Hold and Wait
C) Preemption
D) Circular Wait

20. What is the difference between a process and a thread?

A) A process is a lightweight version of a thread
B) A thread is an independent execution unit within a process, sharing the process's memory space
C) Threads have separate memory spaces from their process
D) There is no difference

21. What is virtual memory?

A) Physical RAM only
B) A memory management technique that gives an application the illusion of a large, contiguous memory using disk space
C) A type of cache memory
D) A backup memory device

22. Which page replacement algorithm suffers from Belady's Anomaly?

A) LRU
B) FIFO
C) Optimal
D) LFU

23. What is a deadlock?

A) A state where two or more processes are waiting indefinitely for resources held by each other
B) A process that has finished execution
C) A type of memory leak
D) A CPU scheduling technique

24. Which of the following synchronization tools is used to prevent race conditions?

A) Compiler
B) Semaphore
C) Scheduler
D) Loader

25. Which memory management technique divides memory into fixed-size blocks?

A) Segmentation
B) Paging
C) Swapping
D) Caching

26. What is a race condition?

A) When multiple processes/threads access shared data concurrently and the outcome depends on timing
B) A process running faster than expected
C) A type of scheduling algorithm
D) A network protocol

Computer Networks

27. What is the purpose of the ARP protocol?

A) To resolve an IP address to a MAC address
B) To resolve a domain name to an IP address
C) To transfer files
D) To send email

28. Which of the following devices operates primarily at the Network layer to forward packets between different networks?

A) Switch
B) Hub
C) Router
D) Repeater

29. What is the maximum number of usable host addresses in a /24 subnet?

A) 256
B) 254
C) 255
D) 512

30. Which of the following best describes a subnet mask?

A) It divides an IP address into network and host portions
B) It encrypts network traffic
C) It assigns MAC addresses
D) It is used only in IPv6

Data Structures & Algorithms

31. Which of the following problems is a classic example solved using dynamic programming?

A) Binary search
B) 0/1 Knapsack problem
C) Bubble sort
D) Linear search

32. What is the time complexity of the Two-Pointer technique on a sorted array to find a pair with a given sum?

A) O(n^2)
B) O(n log n)
C) O(n)
D) O(log n)

33. Which data structure uses LIFO (Last In First Out) order?

A) Queue
B) Stack
C) Linked List
D) Array

34. What is the space complexity of an iterative approach to computing Fibonacci numbers with memoization storing all n results?

A) O(1)
B) O(n)
C) O(n^2)
D) O(log n)

35. Which data structure is typically used to implement DFS (iteratively)?

A) Queue
B) Stack
C) Heap
D) Hash table

36. What is tail recursion?

A) A recursive call that is the last operation in a function, which can be optimized by some compilers into iteration
B) A recursive function with no base case
C) A function that calls itself twice
D) A type of infinite loop

37. Which of the following is a common technique to resolve hash collisions?

A) Chaining or open addressing
B) Binary search
C) Recursion
D) Memoization

38. What is the time complexity of binary search on a sorted array of n elements?

A) O(n)
B) O(log n)
C) O(n log n)
D) O(1)

39. What is dynamic programming primarily used for?

A) Solving problems by breaking them into overlapping subproblems and storing results to avoid recomputation
B) Sorting arrays only
C) Random number generation
D) Compiling code faster

40. What is a hash collision?

A) When two different keys map to the same hash bucket/index
B) When a hash table runs out of memory
C) When a key is deleted incorrectly
D) When two hash tables merge

41. What is the worst-case time complexity of QuickSort?

A) O(n log n)
B) O(n^2)
C) O(log n)
D) O(n)

42. What is the amortized time complexity of appending an element to a dynamic array (like Python's list or Java's ArrayList)?

A) O(n)
B) O(log n)
C) O(1) amortized
D) O(n^2)

43. What is a balanced binary search tree used to prevent?

A) Overflow errors
B) Degeneration into a linked list (worst-case O(n) operations)
C) Memory leaks
D) Integer overflow

44. Which data structure uses FIFO (First In First Out) order?

A) Stack
B) Queue
C) Tree
D) Graph

45. Which data structure underlies the implementation of recursion internally?

A) Queue
B) Stack (call stack)
C) Heap (memory heap)
D) Graph

46. What is the time complexity of Bubble Sort in the worst case?

A) O(n)
B) O(n log n)
C) O(n^2)
D) O(log n)

47. What does BFS stand for in graph traversal?

A) Best First Search
B) Breadth First Search
C) Binary First Search
D) Backtrack First Search

48. Which algorithm is used to find the Minimum Spanning Tree of a graph?

A) Dijkstra's Algorithm
B) Kruskal's or Prim's Algorithm
C) Bellman-Ford Algorithm
D) Floyd-Warshall Algorithm

49. What is the time complexity of inserting an element at the head of a singly linked list?

A) O(n)
B) O(1)
C) O(log n)
D) O(n log n)

50. Which of the following is true about a doubly linked list compared to a singly linked list?

A) It uses less memory per node
B) Each node has pointers to both the next and previous nodes, allowing traversal in both directions
C) It cannot be traversed backward
D) It has no advantages

51. What is the time complexity of Floyd-Warshall algorithm for all-pairs shortest paths?

A) O(V^2)
B) O(V^3)
C) O(E log V)
D) O(V + E)

52. Which algorithmic technique does Merge Sort primarily use?

A) Greedy
B) Divide and Conquer
C) Dynamic Programming
D) Backtracking

53. What is the average time complexity of searching in a well-implemented hash table?

A) O(n)
B) O(log n)
C) O(1)
D) O(n^2)

54. Which of the following best describes a trie data structure?

A) A tree-like structure used for efficient retrieval of strings, especially for prefix matching
B) A type of balanced binary search tree
C) A hashing algorithm
D) A graph traversal method

55. What is the primary difference between a greedy algorithm and dynamic programming?

A) Greedy makes a locally optimal choice at each step without reconsidering, DP considers overlapping subproblems and often guarantees a globally optimal solution
B) They are the same technique
C) DP never uses recursion
D) Greedy always gives an optimal solution while DP does not

56. Which data structure is typically used to implement BFS?

A) Stack
B) Queue
C) Heap
D) Array only

57. Which sorting algorithm is considered stable and has O(n log n) time complexity in all cases?

A) QuickSort
B) Merge Sort
C) Selection Sort
D) Heap Sort

58. Which technique explores all possible options and backtracks when a solution path fails, often used in puzzles like N-Queens?

A) Greedy algorithm
B) Backtracking
C) Dynamic programming
D) Binary search

59. What is the time complexity of building a heap from an unsorted array (heapify)?

A) O(n log n)
B) O(n)
C) O(log n)
D) O(n^2)

60. What is a monotonic stack typically used for?

A) Finding the next greater/smaller element efficiently
B) Sorting an array in place
C) Implementing recursion
D) Hashing

61. What is the time complexity of Dijkstra's algorithm using a binary heap?

A) O(V^2)
B) O(E log V)
C) O(V + E)
D) O(E^2)

62. What is the time complexity of accessing an element in an array by index?

A) O(n)
B) O(log n)
C) O(1)
D) O(n log n)

63. Which traversal of a Binary Search Tree gives elements in sorted order?

A) Pre-order
B) Post-order
C) In-order
D) Level-order

64. What is memoization?

A) Storing the results of expensive function calls and reusing them when the same inputs occur again
B) A sorting technique
C) A type of recursion without a base case
D) A method to delete duplicate array elements

65. Which data structure is best suited for implementing a priority queue efficiently?

A) Array
B) Linked List
C) Heap
D) Stack

66. Which of these best describes the 'sliding window' technique?

A) Maintaining a subset of elements (a window) over a sequence and adjusting its bounds to solve subarray/substring problems efficiently
B) A sorting technique for linked lists
C) A hashing technique
D) A tree traversal method

Programming Language Fundamentals

67. What is the purpose of the 'finally' block in exception handling (Java/Python)?

A) It executes only if an exception occurs
B) It executes only if no exception occurs
C) It always executes regardless of whether an exception occurred or not
D) It is used to declare exceptions

68. Which of the following is true about Python's Global Interpreter Lock (GIL)?

A) It allows true parallel execution of threads on multiple cores
B) It ensures only one thread executes Python bytecode at a time in CPython, limiting CPU-bound multithreading performance
C) It has no impact on multithreading
D) It is used only in Python 2

69. Which keyword is used to handle exceptions in Python?

A) catch
B) except
C) rescue
D) handle

70. What is the purpose of a 'static' variable in a class (Java/C++)?

A) It belongs to the instance and is unique per object
B) It belongs to the class rather than any instance and is shared across all instances
C) It can only be used inside loops
D) It is always private

71. What does 'immutable' mean for an object (e.g., a String in Java)?

A) Its value can be changed after creation
B) Its state cannot be changed after it is created
C) It can only be used once
D) It cannot be passed to functions

72. What is duck typing (commonly associated with Python)?

A) A type system that checks types strictly at compile time
B) An object's suitability is determined by the presence of certain methods/behavior rather than its explicit type
C) A type of exception
D) A memory allocation strategy

73. In Java, what is autoboxing?

A) Automatic conversion between primitive types and their corresponding wrapper classes
B) A way to box multiple objects together
C) A garbage collection technique
D) A type of casting error

74. What is the difference between a shallow copy and a deep copy of a list containing nested lists?

A) A shallow copy also copies nested objects recursively, deep copy does not
B) A shallow copy copies references to nested objects, deep copy recursively copies nested objects too
C) They behave identically
D) Deep copy is not possible in most languages

75. What is the difference between a compiler and an interpreter?

A) A compiler translates the whole program to machine code before execution, an interpreter executes code line-by-line

B) They are identical

C) Interpreters are always faster than compilers

D) Compilers cannot detect syntax errors

76. In Java, what is the difference between '==' and '.equals()' when comparing objects?

A) They always behave identically

B) '==' compares references while '.equals()' compares content (if overridden)

C) '.equals()' is only for primitives

D) '==' is only used for Strings

77. What is a lambda function/expression?

A) A named function stored in a file

B) An anonymous, inline function typically used for short operations

C) A type of class

D) A compiler directive

78. What is method chaining?

A) Calling multiple methods on the same object in a single statement by having each method return the object itself

B) A way to declare multiple classes

C) A type of exception handling

D) A memory management technique

79. In C++, what does the 'const' keyword do when applied to a variable?

A) Makes the variable a pointer

B) Prevents the variable's value from being modified after initialization

C) Declares a static variable

D) Declares a global variable

80. Which of the following best describes garbage collection?

A) A manual process where the developer frees memory explicitly

B) An automatic process that reclaims memory occupied by objects no longer in use

C) A compilation step

D) A type of exception handling

Web & Cloud Fundamentals

81. What does CSS stand for?

A) Cascading Style Sheets

B) Computer Style Sheets

C) Creative Style System

D) Cascading Software Sheets

82. What does HTTP stand for?

A) HyperText Transfer Protocol
B) HighText Transmission Protocol
C) Hyperlink Transfer Text Protocol
D) HyperText Trade Protocol

83. What is auto-scaling in cloud computing?

A) Automatically adjusting compute resources up or down based on demand
B) Manually adding servers only
C) A type of database indexing
D) A billing model only

84. What is the primary difference between HTTP and HTTPS?

A) HTTPS uses encryption (SSL/TLS) to secure data in transit while HTTP does not
B) HTTP is faster in all cases
C) HTTPS cannot use port 443
D) There is no difference

85. What is the purpose of a Content Delivery Network (CDN)?

A) To cache and serve content from servers geographically closer to users, reducing latency
B) To encrypt all web traffic
C) To replace DNS servers
D) To manage database transactions

86. What is the difference between IaaS, PaaS, and SaaS?

A) They are all the same cloud service model
B) IaaS provides raw infrastructure, PaaS provides a platform to build apps, SaaS delivers ready-to-use software
C) SaaS provides infrastructure while IaaS provides software
D) PaaS is only for storage

87. What is the purpose of cookies in web applications?

A) To store small pieces of data on the client side to maintain state/session information
B) To encrypt HTTPS traffic
C) To speed up DNS resolution
D) To compress images

88. What is a RESTful API?

A) An API that strictly requires SOAP protocol
B) An architectural style for designing networked applications using stateless operations over standard HTTP methods
C) A type of database
D) A programming language

89. What does 'statelessness' mean in the context of REST APIs?

A) The server stores client session state between requests
B) Each request from a client contains all the information needed to process it, with no reliance on stored server-side session state
C) The API has no state at all, including no data
D) Stateless APIs cannot use authentication

90. Which HTTP method is typically used to retrieve data without side effects?

A) POST

B) GET

C) DELETE

D) PUT

System Design Basics

91. What is the CAP theorem in distributed systems?

A) A system can simultaneously guarantee Consistency, Availability, and Partition tolerance

B) A distributed system can only guarantee two of Consistency, Availability, and Partition tolerance at the same time during a network partition

C) A theorem about compiler design

D) A theorem unrelated to distributed systems

92. What is horizontal scaling?

A) Adding more power (CPU/RAM) to an existing server

B) Adding more servers/machines to distribute the load

C) Reducing the number of servers

D) Only applicable to databases

93. What is the purpose of a message queue (e.g., Kafka, RabbitMQ) in system design?

A) To decouple producers and consumers of data, enabling asynchronous processing and buffering

B) To replace a database

C) To render a webpage

D) To manage user sessions only

94. What is idempotency in the context of API design?

A) An operation that produces the same result no matter how many times it is performed

B) An operation that can never be repeated

C) An operation that always fails on retry

D) A synonym for authentication

95. What is a load balancer used for in system design?

A) To store data persistently

B) To distribute incoming requests across multiple servers to avoid overload on any single server

C) To compress images

D) To manage user authentication only

96. What is eventual consistency?

A) A guarantee that all replicas are updated instantly

B) A consistency model where, given enough time without new updates, all replicas converge to the same value

C) A model where data is never consistent

D) A synonym for strong consistency

97. What is the purpose of an API Gateway in a microservices architecture?

A) To act as a single entry point that routes requests to appropriate backend services and can handle cross-cutting concerns like auth and rate limiting

B) To store all application data

C) To replace load balancers entirely

D) To compile microservices code

98. What is vertical scaling?

A) Adding more machines to a system

B) Increasing the resources (CPU/RAM) of an existing machine

C) Distributing data across regions

D) A type of caching

99. What is caching used for in system design?

A) To permanently store all application data

B) To temporarily store frequently accessed data closer to the application for faster retrieval

C) To replace a database entirely

D) To handle user authentication

100. What is database sharding used for?

A) To horizontally partition a large database across multiple servers to improve scalability

B) To encrypt a database

C) To back up data

D) To normalize tables

Answer Key

1-B	2-C	3-A	4-A	5-B	6-A	7-C	8-B	9-A	10-A
11-D	12-B	13-A	14-B	15-C	16-B	17-B	18-A	19-C	20-B
21-B	22-B	23-A	24-B	25-B	26-A	27-A	28-C	29-B	30-A
31-B	32-C	33-B	34-B	35-B	36-A	37-A	38-B	39-A	40-A
41-B	42-C	43-B	44-B	45-B	46-C	47-B	48-B	49-B	50-B
51-B	52-B	53-C	54-A	55-A	56-B	57-B	58-B	59-B	60-A
61-B	62-C	63-C	64-A	65-C	66-A	67-C	68-B	69-B	70-B
71-B	72-B	73-A	74-B	75-A	76-B	77-B	78-A	79-B	80-B
81-A	82-A	83-A	84-A	85-A	86-B	87-A	88-B	89-B	90-B
91-B	92-B	93-A	94-A	95-B	96-B	97-A	98-B	99-B	100-A

Format: [Question No.]-[Correct Option Letter]. Content compiled from publicly available 2026 placement-prep resources and established CS-fundamentals question banks - verify current-year specifics before your interview.